

APPLIED DATA SCIENCE
CAPSTONE PROJECT
OKO ALEXANDER
TYLER KELLY
THOMPSON KEMP
AUSTIN CALDWELL

Project Description

The goal of this project is to apply the knowledge gained through the Data Science program to solve real-world interdisciplinary problems. Application areas include computational semantics, automated software analytics, and Data Visualization for Pattern Discovery. Through this project, we aim to critically analyze our findings and derive human-interpretable insights about the phenomena encoded in the data. Our group will be specifically interested in solving this specific problem: How effectively can a large language model be steered using an embedded language model trained on the relational connection between hypothesis and conclusion?

A problem like that generates other questions we would like to answer, such as:

- How accurately can the model associate a hypothesis with its corresponding conclusion?
 - Plan to divide papers into sections, including a section with hypothesis/proposed methodology, general content with results of the experiment, and finally a conclusion section that our Qwen model is attempting to predict. Various cosine similarity methods may be used to determine accuracy.
- Does performance improve as the amount of scientific literature used for training increases?
 - Will test the model with progressively higher volumes of papers with a definite hypothesis and conclusion. The volume of papers will increase based on our confidence in a created model.
- What semantic patterns distinguish hypotheses that lead to different types of conclusions?
 - Test different types of semantic patterns in the exploratory data analysis phase and compare results.
- Are the relationships learned by the model human-interpretable?
 - Can analyze the content of a generated conclusion based on a hypothesis to determine the extent of human interpretability.

We are specifically interested in hypothesis-to-correct-conclusion retrieval: Given a hypothesis, choose the correct conclusion from several conclusions. An extension of this problem could be; how much information about a scientific study's conclusion is encoded in its hypothesis and can a language model learn this relationship?

Relevant Research:

<https://arxiv.org/pdf/2312.06681>

This paper goes over steering Llama 2 using contrastive activation addition. In our current approach, we use this technique of adding a vector with encoded directional meaning during an

LLMs forward pass. In simple terms, we're pushing the LLM's output towards a directional meaning.

<https://arxiv.org/pdf/2406.00024>

This paper is extremely important for our project as it introduces E.A.G.L.E (embedding-aligned guided language). Essentially, we can treat an LLM's latent spaces as the environment and then train an agent that can iteratively steer an LLM's generation towards regions of the embedded space. (In our case, our embedded hypothesis-conclusion pairs)

Toy Pipeline:

1. Construct a dataset of hypotheses and conclusions. Hypotheses act as the input and conclusions as the output/target variable.
2. Train BAAI/bge-m3 (the embedded text model). Here, we use a contrastive loss function to pull similar data points together and push dissimilar ones apart, in order to begin pulling our hypotheses and conclusions together.
3. Calculate the mean translation vector. NOTE: As of now, we are not using EAGLE, and instead are opting for just Inference-Time Activation Steering. Simply add the vector into the forward pass of the LLM.
4. Load the LLM and start generating output tokens. Attach the forward hook (the function that adds our mean vector) to a layer of the model, currently in the middle. Observe model behavior.

Outline For Phase 2:

Phase 2 of this project is for exploratory data analysis; it is here where we will begin to experiment with our steered model to see what changes we can make to improve its effectiveness. This will involve data cleaning to remove things such as outliers, which may affect the steering negatively. We will also make several changes to our current toy pipeline, such as the amount of information on the input, the steering method we use, and the degree to which we steer. This, in conjunction with manipulating the dataset to get cleaner outputs, should help us better answer our questions and potentially discover questions we had not previously thought.

Member Responsibility:

Austin - Data Scraping

Tyler - Pipeline Architecture

Thompson - Data Exploration and Visualization

Alex - Evaluation and Documentation